

Strong Lensing Redshift Distribution

VLT/X-shooter Spectroscopic Observations of DESI Candidates

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I. Gravitational Lensing & Cosmology

- Dark Matter Halos & Dark Energy Constraints
- Deflector Mass Profiles & Kinematics

II. VLT/X-shooter Extraction

- 2D Trace & Spatial Collapse
- Sky Subtraction Protocol

III. Methodological Pitfalls

- Detector Artifacts vs. Emission Lines
- Order Merging Discontinuities

IV. Results & Confirmation

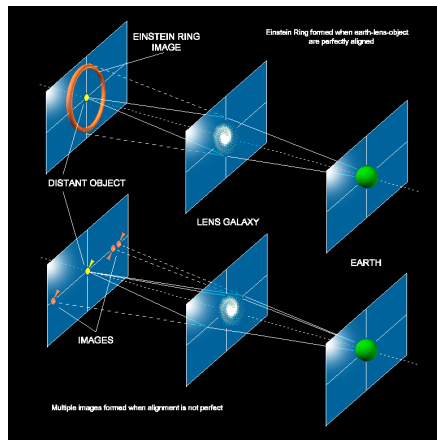
- Deflector Redshifts
- Velocity Dispersions (σ_v) & Mass Modeling

1. Scientific Context & Lensing Theory

Scientific Context: Early-Type Galaxies & Strong Lensing

Probing Mass Distributions with Strong Lensing

- **Early-Type Galaxies (ETGs):** Show complex evolutionary histories (size growth, active SMBH feedback).
- **Strong Lensing Paradigm:** Provides a direct, model-independent measurement of total (luminous + dark) mass.
- **Key Requirement:** Accurate spectroscopic redshifts z_{lens} and z_{source} are needed to convert optical geometry into physical masses.



Lensing geometry and Einstein ring

Einstein Radius & Critical Surface Density

For a source at z_{source} and a deflector lens at z_{lens} , the **Einstein radius** θ_E scales with the angular diameter distances D_A :

$$\theta_E = \sqrt{\frac{4GM(\leq \theta_E)}{c^2} \frac{D_{ls}}{D_{ol}D_{os}}}$$

Critical Surface Density Σ_c :

$$\Sigma_c = \frac{c^2}{4\pi G} \frac{D_{os}}{D_{ol}D_{ls}}$$

Multiple images and arcs form when convergence $\kappa = \Sigma/\Sigma_c \geq 1$.

Total Enclosed Mass $M(\leq \theta_E)$:

$$M(\leq \theta_E) = \frac{c^2}{4G} \frac{D_{ol}D_{os}}{D_{ls}} \theta_E^2$$

Determining accurate distances $D(z)$ strictly requires precise z_{spec} .

Spectral Diagnostics for Lenses and Sources

Foreground Deflectors ($z_{\text{lens}} \sim 0.2 - 0.8$)

- Passive, evolved stellar populations.
- D_n4000 **break** and Balmer break ($\sim 3646 \text{ \AA}$).
- Absorption features: **[CaII] H & K** (3934, 3968 $\text{\AA}$), **Mg b triplet** (5172 $\text{\AA}$).

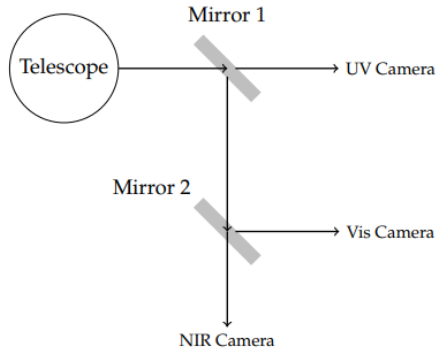
Background Sources ($z_{\text{source}} \sim 1.0 - 3.5$)

- Star-forming, blue lensed arcs.
- Rest-UV continuum drops: **Lyman break** (912 $\text{\AA}$).
- Narrow emission lines: **Ly- α** (1216 $\text{\AA}$), **[OII]** (3727 $\text{\AA}$), **[OIII]** (5007 $\text{\AA}$), **H- α** .

2. Instrumentation & Data Reduction Pipeline

Instrumentation: VLT / X-shooter Spectrograph

- **Instrument:** Wide-band, intermediate resolution ($R \sim 4000 - 18000$) cross-dispersed Echelle spectrograph at VLT UT2.
- **3-Arm Wavelength Splitting:**
 - **UVB:** 300 – 550 nm
 - **VIS:** 550 – 1020 nm
 - **NIR:** 1000 – 2480 nm
- **Advantage:** Simultaneous UV-to-NIR coverage prevents cross-band slit losses and captures features across wide redshift ranges.



X-shooter optical path and dichroic splitting

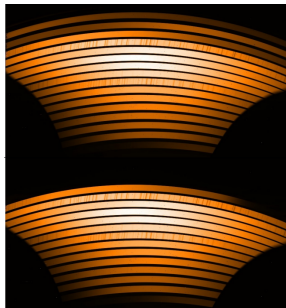
Data Calibration Pipeline & Equation

Master Calibration Frames

Raw science frames are corrected for readout noise, dark current, and pixel sensitivity variations using master Bias, Dark, and Flat-field frames:

$$I_{\text{red}} = \frac{I_{\text{raw}} - D_{\text{master}}}{\left(\frac{F_{\text{master}}}{t_{\text{flat}}} \right) \cdot t_{\text{sci}}}$$

- Data retrieved batch-wise from the ESO Science Archive.
- Transferred via secure VPN/WinSCP to the LAM computing cluster.
- Dithered observations were applied for background sky rejection.



ON-OFF Flat field frames for the normal 0.9" NIR slit (top) and for the 0.9" with blocking filter (bottom). One can easily note that the last orders are cut by the K-band blocking filter.

Raw 2D Echelle orders in the NIR arm

3. Interactive Spectral Analysis & Quality Control

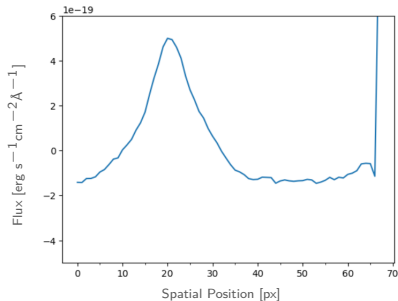
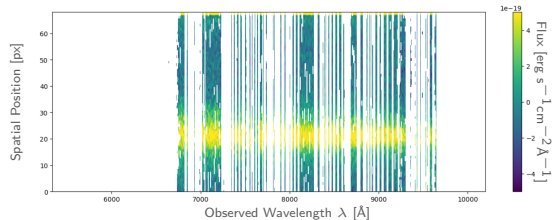
Interactive Extraction Workflow (Jupyter Notebook)

1. Spatial Trace Evaluation

- Collapsing 2D frames along the dispersion axis to pinpoint target location (pixels 15–25).
- Defining off-source sky subtraction regions (pixels 40–60).

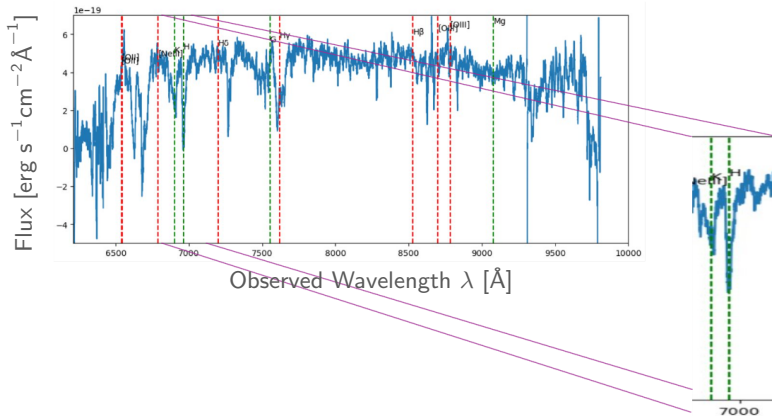
2. Redshift Fitting Protocol

- Locate spectral break (D_n4000 / Balmer).
- Cross-match rest-frame atomic templates.
- Verify consistency across UVB, VIS, and NIR.



2D spectrum and flux profile (DESI-315.3648-43.8104)

1D Spectral Extraction Example: Lens Identification

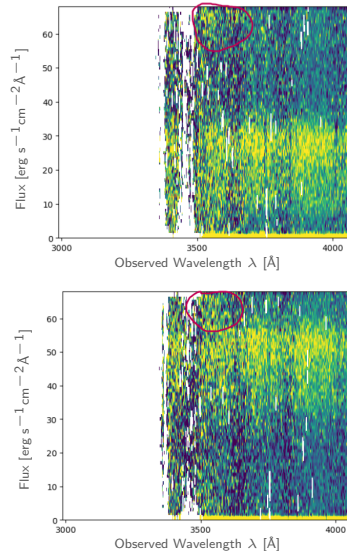


Deflector Redshift Confirmation

Clear alignment of H & K absorption doublet ($3933.7, 3968.5 \text{ \AA}$) yields a secure lens redshift of $z_{\text{lens}} = 0.7545$ for DESI-315.3648-43.8104.

Methodological Pitfall 1: Disambiguating Detector Artifacts

- **False Positive Identification:** A localized flux spot (pixels 40–55) initially mimicked a Ly- α emission line at $z \approx 1.85$.
- **Dithered Frame Check:** Evaluating independent exposures showed that the anomaly **remained stationary on the detector** despite the target nodding shift.
- **Conclusion:** Persistent hot pixel / detector artifact, successfully flagged during spatial quality control.

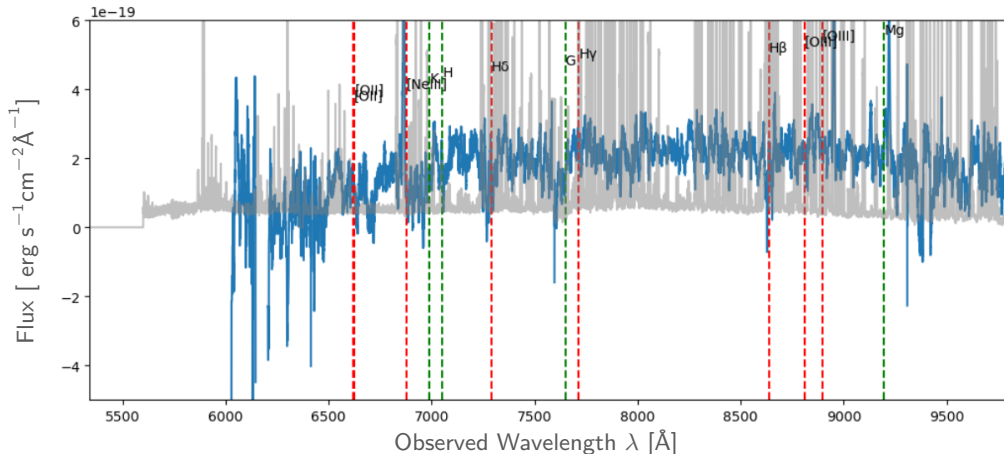


False positive emission line (top) vs dither validation frame (bottom)

Methodological Pitfall 2: Atmospheric Sky Lines

Atmospheric Sky Line Overlay

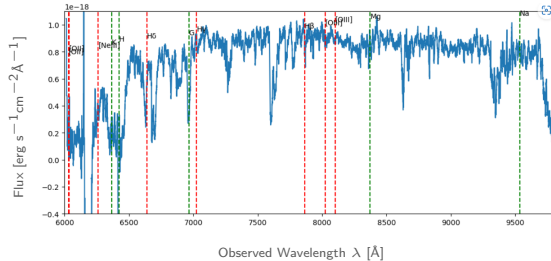
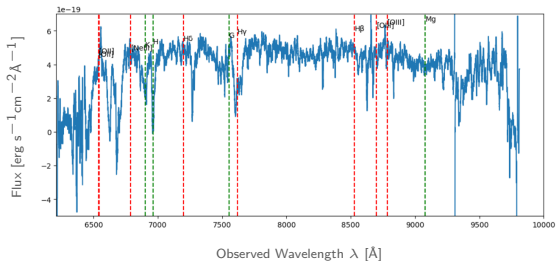
Overlaying synthetic OH atmospheric sky-emission baselines verifies that $z \approx 0.68$ solutions correspond to intrinsic stellar absorption rather than residual atmospheric sky lines.



Methodological Pitfall 2: Order Merging Artifacts

Order Boundary Discontinuities

Imperfect stitching between adjacent Echelle orders introduces artificial continuum jumps, preventing false redshift clustering ($\Delta z \sim 10^{-2}$).



Comparison of extracted 1D optical spectra of candidates DESI-315.3648-43.8104 (left) and DESI-038.8136-21.5095 (right), showing recurring continuum discontinuities caused by order-overlap artifacts rather than intrinsic spectral transitions.

4. Results, Discussion & Perspectives

Summary of Spectroscopic Redshift Measurements

Object ID	z_{source}	z_{lens}	Observational Remarks
DESI-315.3648-43.8104	—	0.7545 ± 0.0010	Single lens redshift identified
DESI-324.5146-60.1293	2.007 ± 0.200	0.6185 ± 0.0020	—
DESI-324.2176-62.6801	—	—	Insufficient SNR
DESI-324.2124-62.6803	—	—	Adverse weather conditions
DESI-016.7705-31.4780	—	0.6855 or 0.7703	Degenerate lens solution
DESI-324.5127-60.1291	2.0185 ± 0.0050	0.6875 or 0.7710 ± 0.001	Degenerate lens solution
DESI-027.4831-28.5753	—	0.6180 ± 0.0090	Uninformative UVB/NIR bands
DESI-015.4984-21.4473	—	0.5175 ± 0.0015	—
DESI-038.8136-21.5095	—	0.6185 ± 0.0010	—
DESI-090.9854-35.9683-B	—	—	Null flux in VIS/UVB; unreduced NIR
DESI-065.6447-28.0652	—	—	Extremely red target; unreduced NIR
DESI-055.0891-25.5584	1.820 ± 0.002	0.6570 ± 0.0010	Robust dual-component extraction
DESI-026.9149-63.5238	—	0.9100 ± 0.0015	High-redshift candidate lens
DESI-355.2727-57.2679	2.1310 ± 0.0160	0.4410 ± 0.0005	—
DESI-338.1418-60.0020	—	0.5710 ± 0.0020	—

Summary of spectroscopic redshift measurements (z_{lens} : deflector galaxy, z_{source} : background lensed object).

Main Findings & Limitations

- Measured mean deflector redshift:
 $\bar{z}_{\text{lens}} \approx 0.67 - 0.71$, between
 $z_{\text{min}} = 0.4410$ and $z_{\text{max}} = 0.91$,
matching galaxy group/cluster lens
scales.
- Low source recovery rate driven by high
atmospheric thermal noise and OH lines
in the NIR arm.

Astrophysical Impact & Next Steps

- Serves as mandatory calibration for mass
modeling combined with Einstein radii
 θ_E .
- Work directly laid the groundwork for
Jullo, Boghossian et al. (2026)
($N = 67$ systems, 58 secure lenses, 57
sources).
- Future cross-matching with space
imaging (Euclid / HST).

- **Data Reduction Routine:** Successfully manipulated a full reduction and interactive extraction workflow for VLT/X-shooter echelle spectra.
- **Quality Control:** Demonstrated that dither checks and sky overlays are essential to remove detector artifacts and order-stitching anomalies.
- **Scientific Contribution:** Securely measured z_{spec} for 7 primary lens galaxies and 4 background lensed sources, providing a valuable spectroscopic calibration for this candidate sample and establishing reliable inputs for dark matter mass profile constraints.

Thank you for your attention!

Technical Appendices

Appendix: X-shooter Optical Layout

- Uses 2 dichroic beam splitters to divide light without energy loss.
- High spectral resolution allows resolving velocity dispersions in massive ETGs ($\sigma_v \sim 200 - 300 \text{ km/s}$).

Arm	Range [Å]	Resolution R (0.9")	Detector	Temp. [K]
UVB	2936 – 5930	$\sim 5\,500$	E2V CCD244-82	153
VIS	5253 – 10489	8 800	MIT/LL CCID-20	135
NIR	9827 – 24807	5 600	Hawaii-2RG	81

Source: Vernet et al. (2011)

Appendix: Detailed X-shooter Optical Scheme

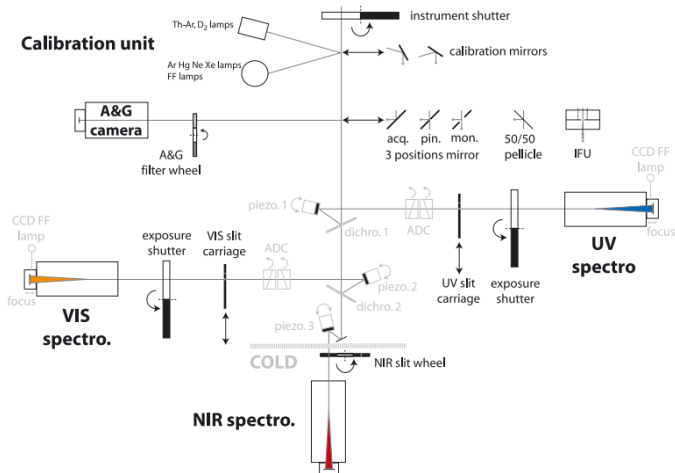


Figure 1: X-shooter optical path and dichroic beam splitting.